

Generative Scientific Discovery Engine

Technical Report

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Abstract

Given 400 sparse, noisy observations of a field $u(x, t)$ and no governing equation, the system retrieves scientific evidence (RAG + knowledge graph), generates candidate equations (LLM + graph templates), tests each with an inverse physics-informed neural network (PINN), rejects invalid hypotheses on held-out data, selects the most parsimonious supported law, and predicts beyond the observed window. It recovers $u_t = D u_{xx}$ with $\hat{D} = 0.10015$ (truth 0.1, error 0.15%), rejects advection and pure reaction, and predicts unseen times with RMSE 1.3×10^{-4} ; out of distribution it is $6\times$ more accurate than a physics-free network. On a second, unseen system it recovers $u_t + c u_x = D u_{xx}$ with c and D within 0.4% and 1.0%. It passes 50/50 benchmark questions, 23/25 multi-hop problems and 20/20 adversarial cases, and the pipeline also runs end to end with a real LLM (Google Gemini).

1 System overview

The pipeline follows the brief: *Observed data* $\rightarrow$ *evidence* $\rightarrow$ *candidate physics* $\rightarrow$ *equation* $\rightarrow$ *PINN* $\rightarrow$ *validation* $\rightarrow$ *new prediction* (Figure 1). Division of responsibility is enforced in code: RAG retrieves, the graph organises, the LLM only *proposes*. Accept/reject and selection use only measured losses on held-out noisy observations; the hidden truth is read in one function, after all decisions, to report error.

Data. Ground truth $u_t = 0.1 u_{xx}$ on $x, t \in [0, 1]$, zero boundary values, $u(x, 0) = \sin \pi x + 0.5 \sin 3\pi x$. The system sees only 400 random points with 5% Gaussian noise. Splits: train on $t \leq 0.8$ (minus a random 20% holdout used for the gate and selection); OOD = all $t > 0.8$, never used for any decision; novel prediction on $t \in (1, 1.5]$, where no data exists. All thresholds were fixed in `config.yaml` before each run.

2 Requirement coverage

Minimum research challenge (all met).

Required	Delivered
≥ 3 generated hypotheses	5 candidates (mock run), 6 (Gemini run)
1 discovered governing equation	$u_t = D u_{xx}$, $\hat{D} = 0.10015$; and $u_t + 0.398 u_x = 0.0198 u_{xx}$ on a second system
1 inverse / 1 forward PINN	Phase 3 (both), every Phase 6 candidate is an inverse PINN
1 sparse/noisy experiment	all main experiments use 400 points with 5% noise
1 OOD prediction	$t > 0.8$: RMSE 0.0022 vs clean field; novel $t \in (1, 1.5]$: 1.3×10^{-4}
1 correctly rejected hypothesis	advection (holdout RMSE 0.227 vs gate 0.03) and pure reaction (0.118)
20 adversarial physics cases	20/20 (3 by abstention)

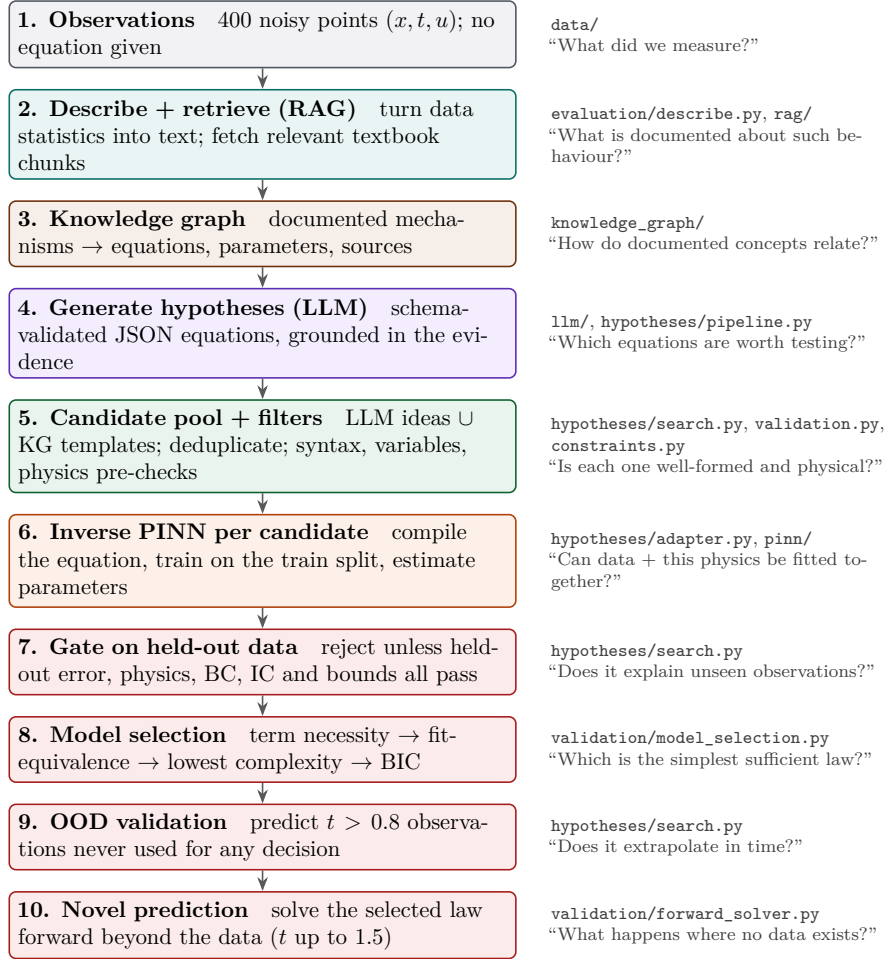


Figure 1: End-to-end pipeline. Colours mark the code package that implements each stage.

Mandatory research components

Mandatory component	Implementation and evidence	Status
Scientific RAG + knowledge graph	<code>rag/</code> , <code>knowledge_graph/</code> ; 22 entities, 40 relationships, all with provenance	Met
Generative hypotheses from multi-source evidence	LLM proposals $\cup$ graph templates (<code>hypotheses/search.py</code>); mock and real Gemini runs	Met
PINN forward and inverse	<code>pinm/</code> ; forward RMSE 7.5×10^{-4} ; inverse $\hat{D}$ error 0.09%	Met
PDE/ODE discovery from data	SINDy (<code>equation_discovery/</code>) on clean/sparse data; PINN-based identification on noisy data	Partial
Unknown parameter estimation (PINN)	D within 0.15%; c , D within 0.4%, 1.0% on a second system	Met
Sparse/noisy-data learning	400 points, 5% noise, two systems	Met
Boundary/initial-condition inference	BC/IC supplied as known physics; enforced and checked, not inferred	Not done
Physics-constrained generative search	bounds ($D, K > 0$), well-posedness pre-checks, PINN gate (Figures 2-3)	Met
Uncertainty estimation	bootstrap coefficient stability (20 subsets); not Bayesian	Partial
Out-of-distribution validation	held-out $t > 0.8$ and novel window $t \in (1, 1.5]$	Met
Comparison with NN and numerical solver	same network without physics; RK4/Euler forward solver	Met

3 Method: physics-constrained generative search

1. **Evidence.** The observations are summarised as text (statistics only, never naming a mechanism) and used to query a 5-document corpus; the knowledge graph returns the documented equations and parameters. A check confirms the corpus never states the dataset’s parameter values (no answer leakage).
2. **Generation.** The LLM returns schema-validated JSON hypotheses; graph templates are added; structural duplicates are removed (parameter- and sign-invariant signatures).
3. **Pre-checks.** Syntax, variables, parameters, declared vs. actual derivatives, unit metadata, evolution form, physical bounds, boundary-condition well-posedness.
4. **Inverse PINN per candidate.** Each equation string is compiled by SymPy into the residual of one shared PINN (4×64 tanh, Adam 3000 + L-BFGS 300); unknown parameters train jointly with the network.
5. **Gate** (held-out noisy data): holdout RMSE ≤ 0.03 , physics residual ≤ 0.01 , BC loss ≤ 0.005 , IC loss ≤ 0.01 , physical bounds.
6. **Selection** (Figure 3): fit-equivalence (within 5% of best) $\rightarrow$ drop inactive terms ($<5\%$ of u_t) $\rightarrow$ lowest complexity $\rightarrow$ BIC.
7. **Validation and prediction.** Bootstrap stability, conservation check, OOD prediction, and a forward solve of the selected law beyond the data.

Ranking criteria requested by the brief, and where each is used:

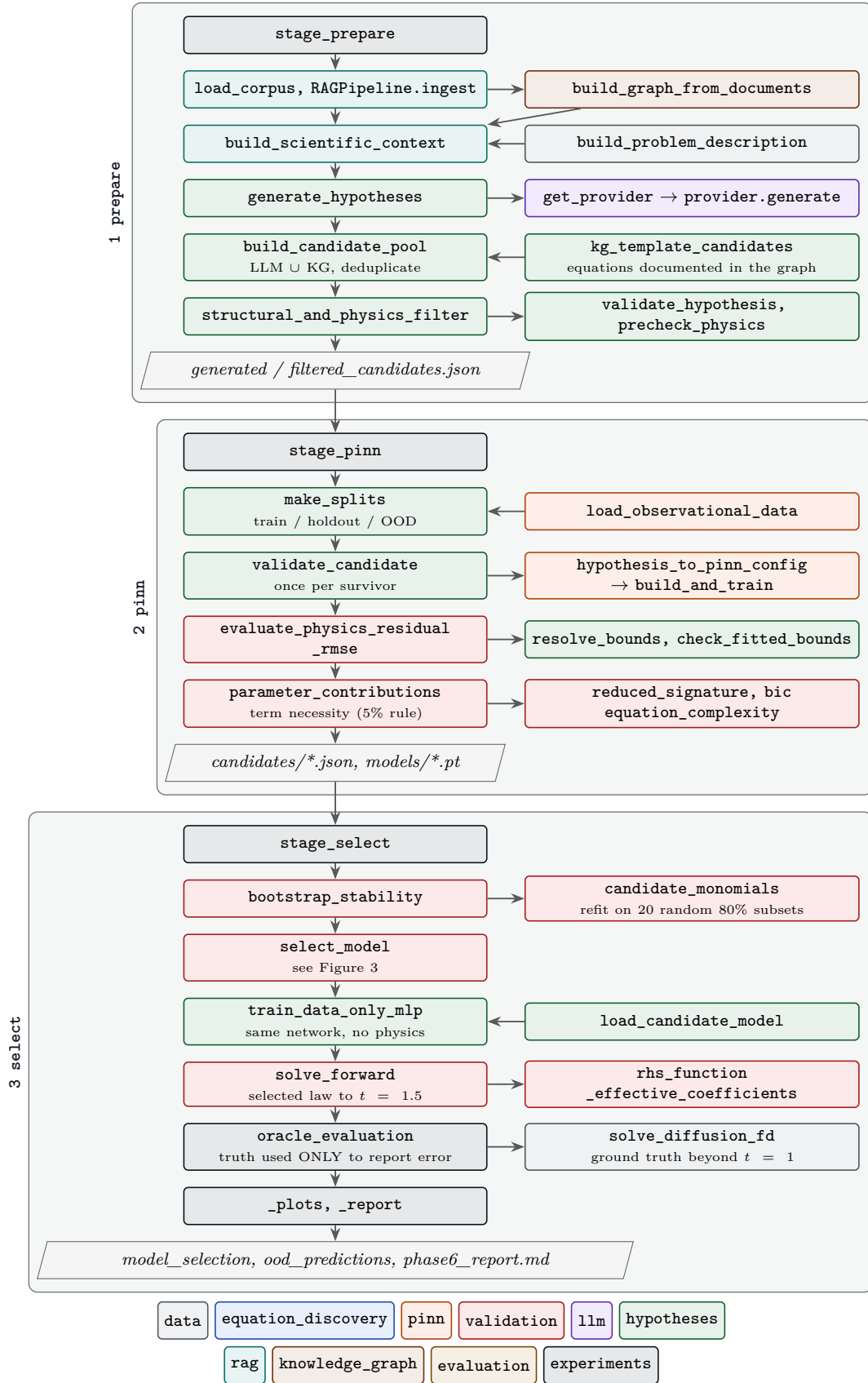


Figure 2: Function-level flow of the search (experiments/08_physics_constrained_search.py), stage by stage. Only oracle_evaluation reads the ground truth, after all decisions.

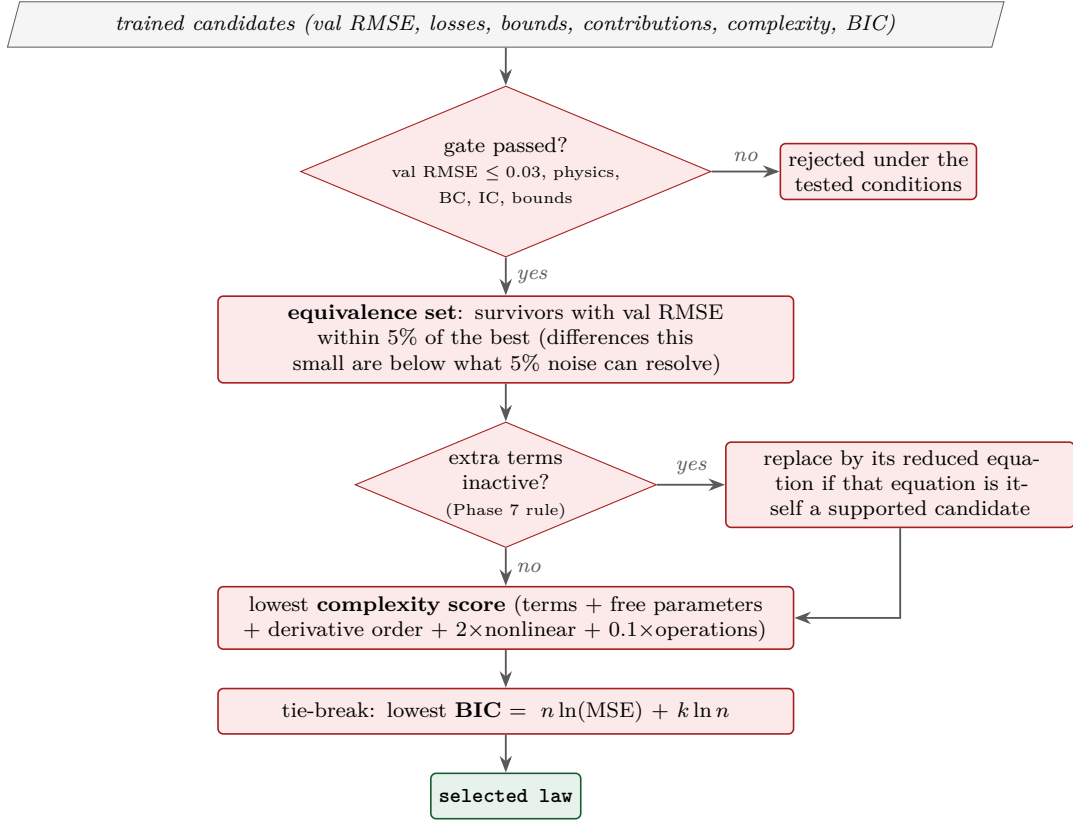


Figure 3: Model-selection rule (`validation/model_selection.py`); thresholds fixed before any run.

Criterion	Measure	Used in
Data fit	holdout RMSE on noisy observations	gate, equivalence
PDE residual	mean squared residual on a fresh grid	gate
Conservation laws	global balance violation $\text{RMS}_t \int \mathcal{R} dx / \text{RMS}_t dM/dt$	reported
Dimensional consistency	declared units per parameter (fraction complete)	pre-check (metadata)
Boundary/initial conditions	BC and IC losses; well-posedness warnings	gate, pre-check
Complexity	terms + parameters + derivative order + 2×nonlinear + 0.1×ops	selection
Uncertainty	bootstrap coefficient mean, spread, sign stability	reported

4 Results

4.1 Hypothesis testing and the discovered law

ID	Candidate	Estimates	Holdout	Status	Inactive	Cons. viol.
H1	$u_t = D u_{xx}$	$D = 0.10015$	0.01215	selected	–	0.10%
H2	$u_t + c u_x = 0$	$c = 0.0082$	0.2267	rejected	–	100%
H3	$u_t = D u_{xx} + r u(1 - u/K)$	$D=0.1008, r=0.019, K=1.18$	0.01212	supported	r, K	0.10%
H4	$u_t + c u_x = D u_{xx}$	$c=-0.0018, D=0.10002$	0.01211	supported	c	0.19%
K3	$u_t = r u(1 - u/K)$ (graph)	$r=-1.43, K=6.98$	0.1169	rejected	–	0.63%

H1, H3 and H4 fit to the noise floor (≈ 0.012); H3 and H4 carry inactive terms and reduce to H1, which has the lowest complexity, and BIC agrees. A pure best-fit rule would have chosen H4 (lower by 0.3%, far below what 5% noise can resolve). Advection breaks global conservation completely: with zero boundary values it conserves $\int u dx$, while the data loses it.

4.2 PINN research results

Metric	Forward (D fixed)	Inverse (D from 0.2)	Phase 6 selected (H1)
PDE residual (fresh grid)	2.8×10^{-5}	2.6×10^{-5}	2.2×10^{-5}
Data loss	1.32×10^{-4}	1.32×10^{-4}	1.29×10^{-4}
Boundary / initial loss	$2.5 \times 10^{-6} / 9.4 \times 10^{-7}$	$2.8 \times 10^{-6} / 7.9 \times 10^{-7}$	$1.3 \times 10^{-6} / 9.0 \times 10^{-7}$
Parameter error	–	0.09% ($\hat{D} = 0.09991$)	0.15% ($\hat{D} = 0.10015$)
Prediction error (vs clean)	RMSE 7.5×10^{-4}	RMSE 8.5×10^{-4}	OOD RMSE 2.2×10^{-3}
Conservation violation	–	–	0.10% (train), 0.37% (OOD), 3.3% ($t > 1$)
Uncertainty (bootstrap u_{xx} coef.)	–	–	0.110 ± 0.003 , sign-stable 100%
OOD (noisy holdout, $t > 0.8$)	–	–	RMSE 0.0116 (noise floor ≈ 0.012)

The bootstrap uses finite-difference derivatives and is biased high (0.110 vs. the PINN’s 0.100); it measures term stability, not a calibrated interval.

4.3 Comparison with a conventional network and a numerical solver

Model	OOD RMSE, $t > 0.8$ (vs clean)	Conservation viol. (train / OOD)
PINN, selected law H1	0.0022	0.10% / 0.37%
Same network, data only (no physics)	0.0139 (6× worse)	14.9% / 5.4%
Forward solver with $\hat{D}$, no data	2.8×10^{-5} ; novel $t \in (1, 1.5]$: 1.3×10^{-4}	–

4.4 Sparse/noisy data and generalisation

SINDy recovers $u_t = 0.1009 u_{xx}$ (clean) and $0.1013 u_{xx}$ (sparse) but fails on the 400 noisy points (five spurious terms); the PINN path succeeds on the same data. On a second hidden system ($u_t + 0.4 u_x = 0.02 u_{xx}$, $x \in [0, 1.5]$, Gaussian start, 400 points, 5% noise) the identical pipeline rejects diffusion, advection and reaction–diffusion and selects advection–diffusion with $c = 0.39841$, $D = 0.01979$ (errors 0.40%, 1.04%), OOD RMSE 0.0013 vs. 0.0066 for the physics-free network (5×), and 0.89% error beyond the data window. SINDy on that data returned a spurious equation.

4.5 Real LLM (Gemini)

With `gemini-3.5-flash-lite` (larger models were overloaded), retrieved evidence changed the generated hypotheses on all three test datasets (for diffusion data, plain diffusion appeared only when evidence was supplied). In the full search Gemini proposed Du_{xx} , $Du_{xx} - ku$, $u_t + cu_x = Du_{xx} - \gamma u$ and $Du_{xx} + ru(1 - u)$; all four passed the gate, but every extra term was inactive, so each reduced to diffusion. The graph added pure advection and pure reaction, which were rejected. The same law was selected ($\hat{D} = 0.10001$, OOD RMSE 0.0020 vs. clean). LLM output is stochastic: a repeated run proposes different alternatives but reaches the same conclusion.

5 Benchmark and ablation

Twelve benchmark datasets (diffusion, advection, reaction–diffusion, advection–diffusion; 3 settings each). **50 questions: 50/50; 25 multi-hop problems: 23/25** (both failures from top-1 lexical retrieval); **20 adversarial cases: 20/20**, including invalid equations, negative diffusivity, time-reversed

data, disguised duplicates, malicious LLM output, a non-identifiable dataset, Occam traps, answer leakage and prompt injection (three noise/sparsity cases pass by abstention).

Ablation arm (12 datasets)	Correct law	OOD within 5%
LLM/RAG generation, no validation	3/12	3/12
RAG + KG generation, no validation	3/12	3/12
Equation discovery alone (SINDy)	8/12	6/12
Full system (generation + physics validation + selection)	12/12	12/12
Full system without the LLM	12/12	12/12

Physics validation and selection drive accuracy; RAG supplies a candidate pool containing the truth (top-1 retrieval only 1/6); the graph and mock LLM add no measurable accuracy here. A leakage audit confirms no decision function reads ground truth and catches a planted cheating selector.

6 Limitations

- Synthetic 1-D PDEs only; known laws recovered, no new law claimed.
- Boundary/initial conditions are given, not inferred.
- Uncertainty is a bootstrap of term stability, not a Bayesian posterior.
- Nested models fit equally well; selection relies on a stated parsimony rule, not uniqueness.
- Lexical retrieval over a 5-document corpus; thresholds hand-set (fixed in advance, not calibrated).
- Conservation is a consistency check, not a discriminator: a network fitted to a wrong law can still follow that law (K3: 0.63%), so it is reported alongside the data gate rather than replacing it.

7 Deliverables and reproducibility

Artifact	Location
Code	data/, equation_discovery/, pinn/, validation/, llm/, hypotheses/, rag/, knowledge_graph/, evaluation/; 95 tests in tests/
Datasets	data/generated/*.npz (clean, sparse, noisy, benchmark, second system)
Trained models	results/phase6/models/, results/evaluation/generalization/models/
Discovered equations	results/phase6/model_selection.json, .../generalization/generalization_results.json
Knowledge graph	results/rag_kg/knowledge_graph.json
Experiments	experiments/02–12 (one script per phase)
Benchmark	results/evaluation/ (questions, multi-hop, adversarial, ablation, leakage)
Conservation; Gemini run	results/evaluation/conservation/; results/evaluation/real_llm_gemini/
Reports	this report (Report.pdf, source docs/Report.tex); results/final_report.md; docs/development_log.md

8 Reproducing these results

All numbers in this report come from one complete run in this environment: macOS (Apple Silicon), CPU only, Python 3.14.7, PyTorch 2.14.0, NumPy 2.5.3, SciPy 1.18.1, pysindy 2.1.0, google-genai 2.28.0. Run every command from the project root (scientific-discovery-engine/), in this order.

1. Install and test (about 25 s):

```
python -m pip install -r requirements.txt
python -m pytest tests -q
```

expect: 95 passed

2. Mock-LLM pipeline. Run these *without* the Gemini key loaded, so the offline mock LLM is used (each script prints a warning that it is falling back to the mock; that is expected).

```
python data/generate_diffusion.py          # 400 noisy points, noise std 0.01159
python experiments/02_equation_discovery.py # SINDy: clean  $u_t = 0.1009 \cdot u_{xx}$ 
python experiments/03_forward_pinn.py      # ~4 min; RMSE vs clean 0.00075
python experiments/04_inverse_pinn.py      # ~4 min;  $D_{\text{hat}} = 0.099909$ 
python experiments/05_hypothesis_validation.py # ~8 min; H1 supported, H2 rejected
python experiments/06_hypothesis_generation.py --skip-pinn
python experiments/07_scientific_rag_kg.py # ~4 min; 22 entities, 40 relations
python experiments/08_physics_constrained_search.py # ~20 min; selects H1,  $D = 0.10015$ 
python experiments/10_generalization.py    # ~16 min; selects H4,  $c = 0.398$ 
python experiments/09_evaluation.py        # ~30 s; 50/50, 23/25, 20/20
```

Script 10 must run before 09, because 09 reads its results.

3. Real-LLM run (Gemini). Put `GEMINI_API_KEY=...` in a `.env` file in the project root, then:

```
set -a; source .env; set +a                # load the key (never printed)
python experiments/11_real_llm_gemini.py --stage generate --model gemini-3.5-flash-lite
python experiments/11_real_llm_gemini.py --stage prepare --model gemini-3.5-flash-lite
python experiments/11_real_llm_gemini.py --stage pinn    # ~25 min; no LLM calls
python experiments/11_real_llm_gemini.py --stage select  # selects H1,  $D = 0.10001$ 
```

This writes only to `results/evaluation/real_llm_gemini/`. With the key loaded, do not re-run scripts 06–08: they would use Gemini and overwrite the mock results.

4. Conservation check (seconds; uses the saved models from steps 2–3):

```
python experiments/12_conservation.py      # H1: 0.10% train, MLP: 14.9%
```

What to expect on another machine. The qualitative results (which equations are accepted, rejected and selected; the benchmark scores) are reproducible. Fitted values and errors can differ slightly between machines and library versions, because neural-network training is sensitive to floating-point details. For example, an earlier run in a different environment gave $\hat{D} = 0.10002$ and a physics-free network that was less accurate. Gemini’s proposals vary from run to run.

A Additional flowcharts

Function-level flows of the remaining components. Colours follow the package legend.

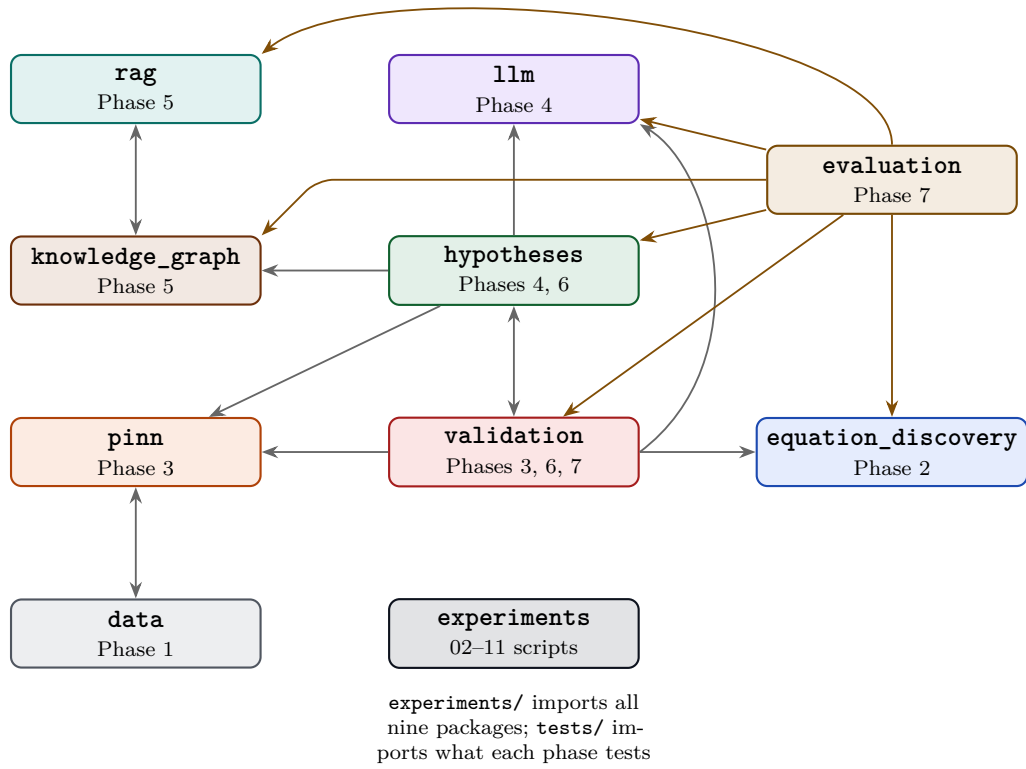


Figure 4: Package dependency map (from the actual imports).

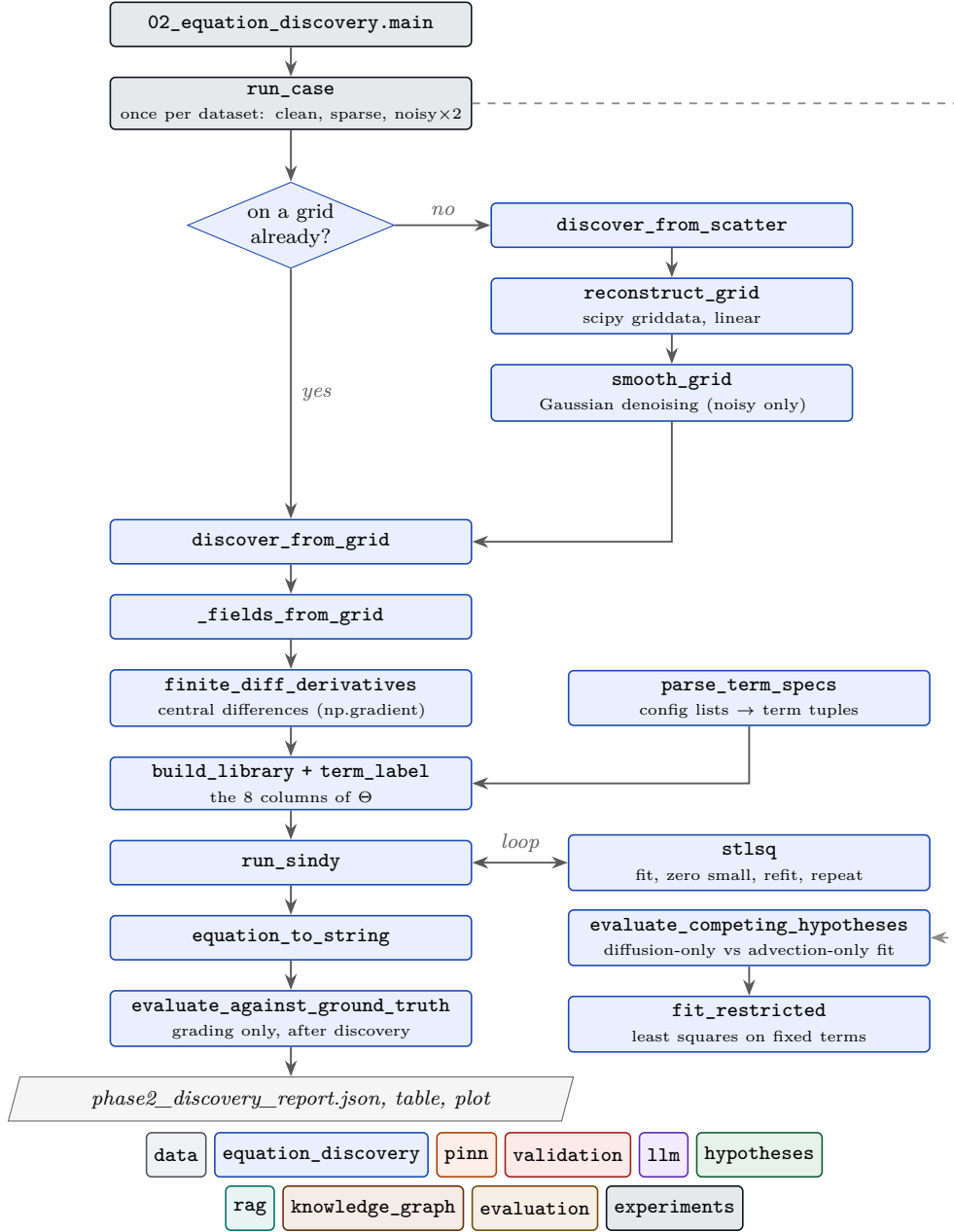


Figure 5: PDE discovery from data (SINDy/STLSQ).

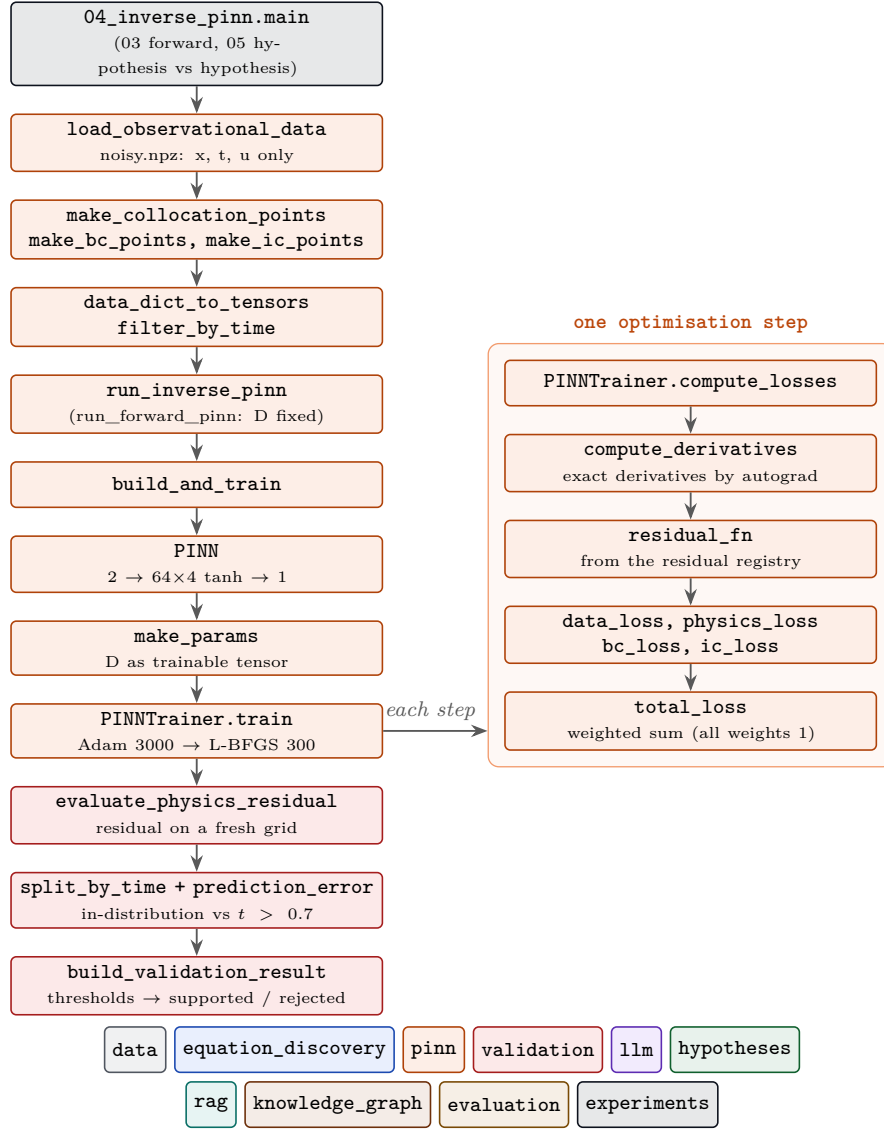


Figure 6: PINN training (forward/inverse) and threshold scoring.

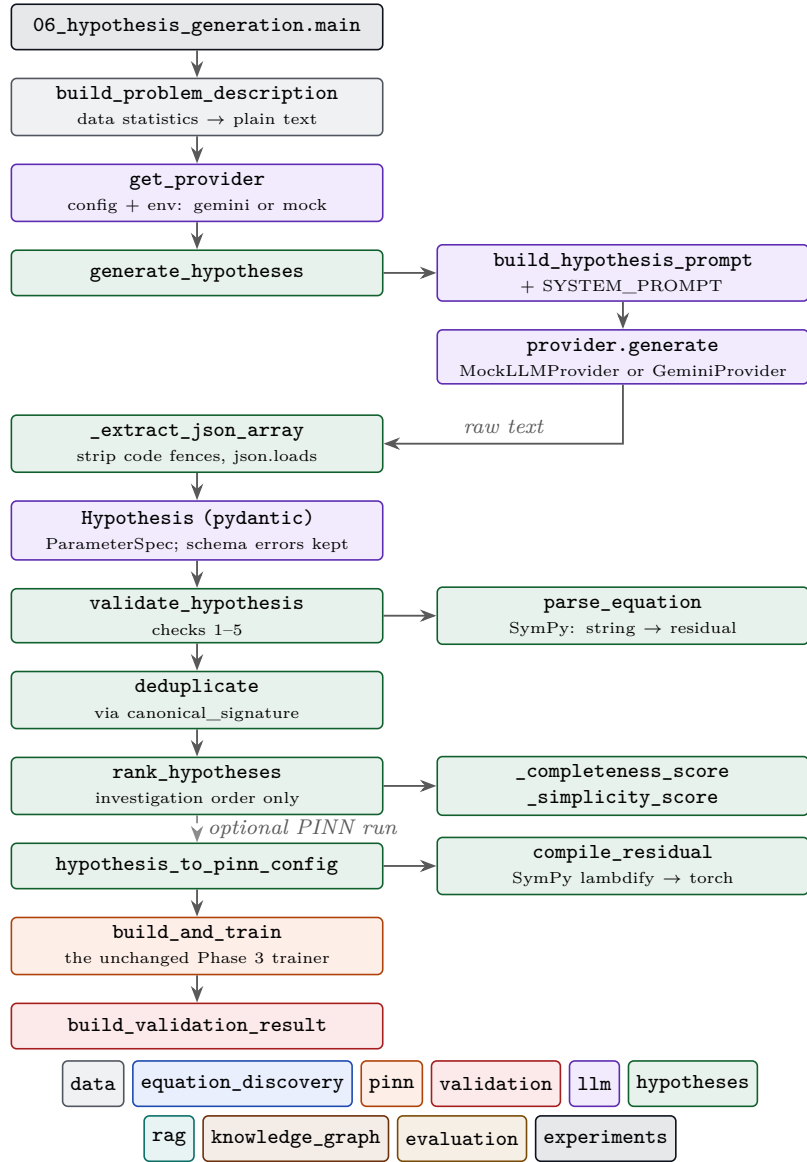


Figure 7: Hypothesis generation, deterministic validation and compilation into a PINN.

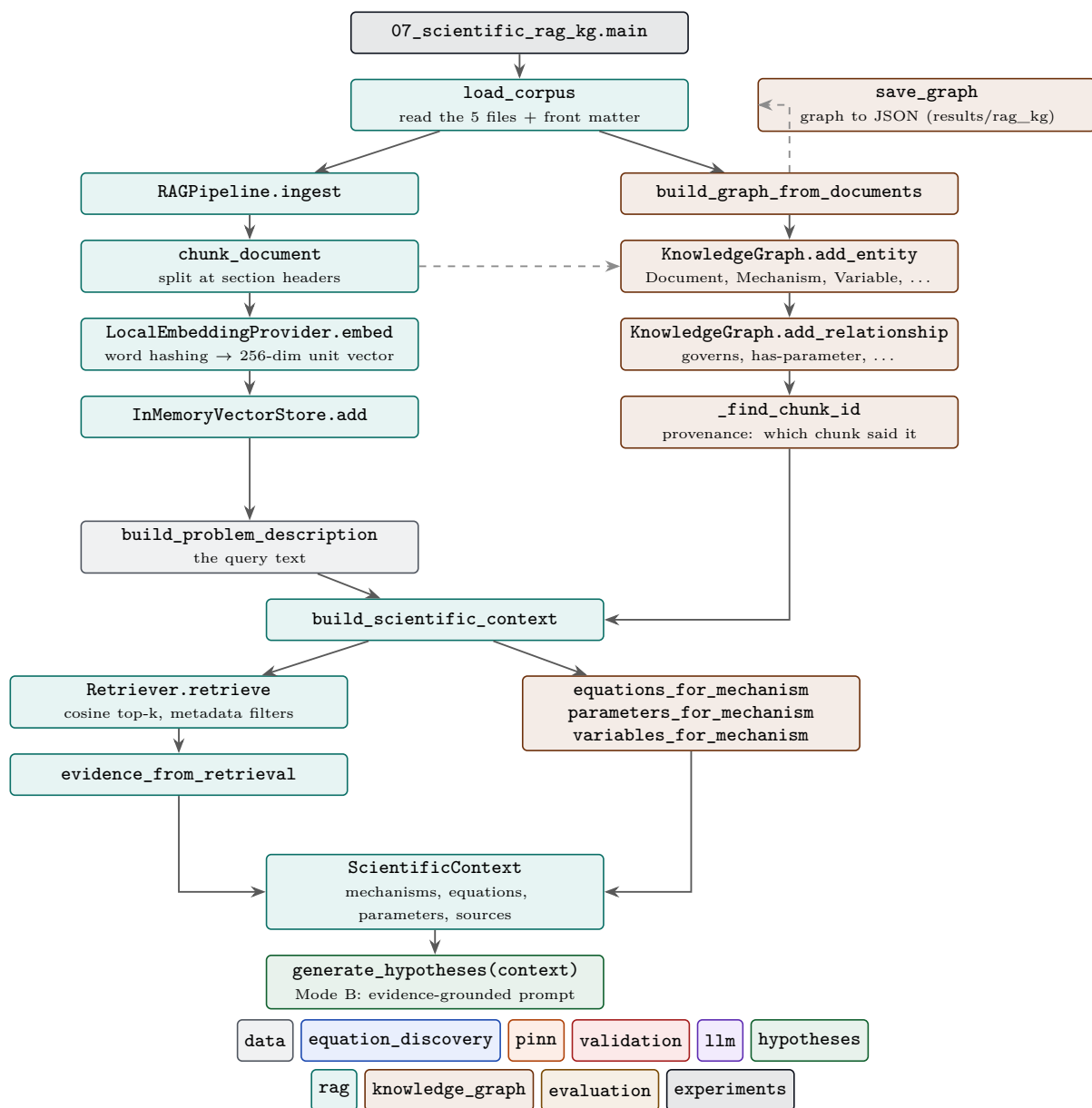


Figure 8: Scientific RAG and knowledge-graph construction into one scientific context.

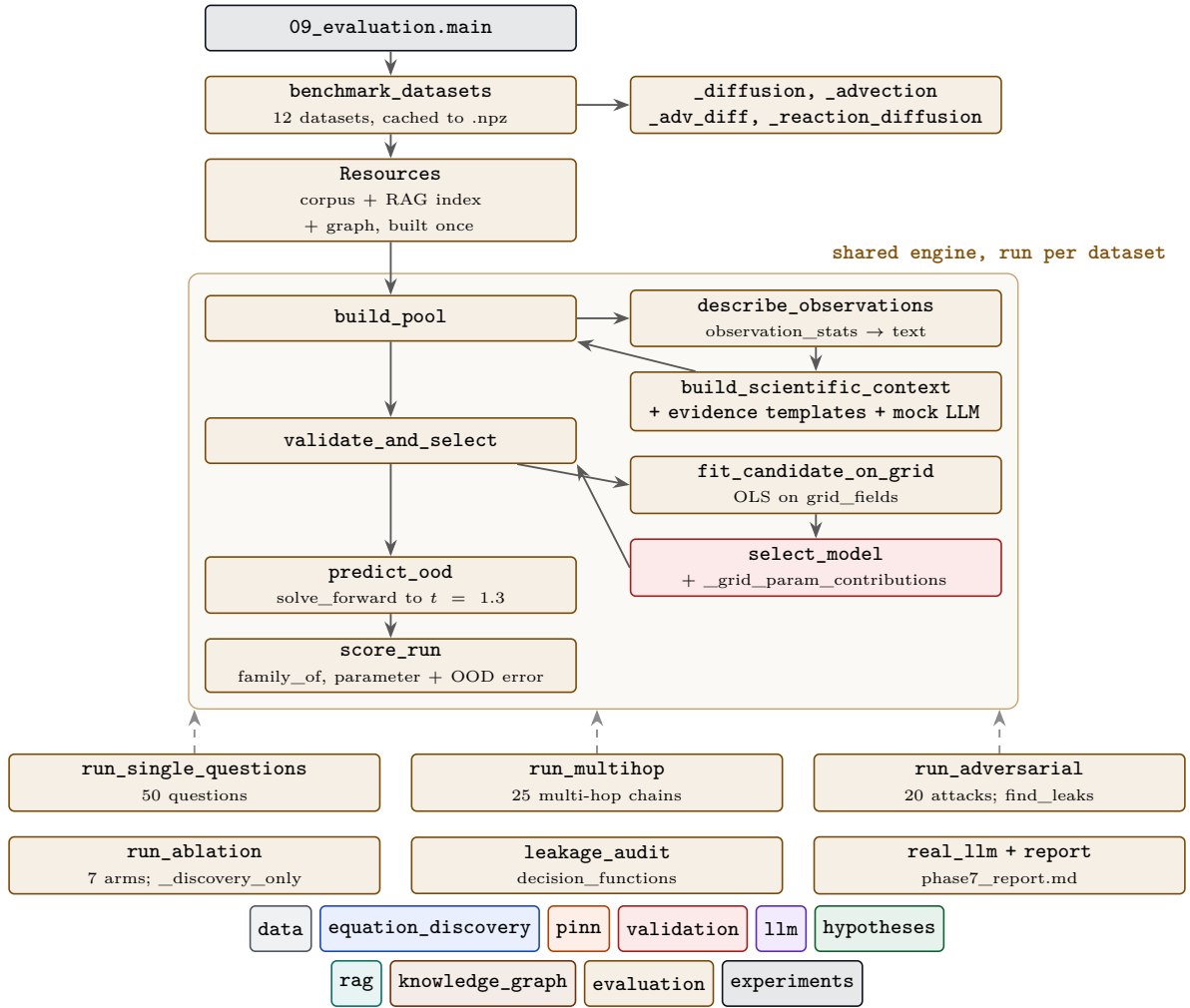


Figure 9: Benchmark, adversarial, ablation and leakage evaluation.

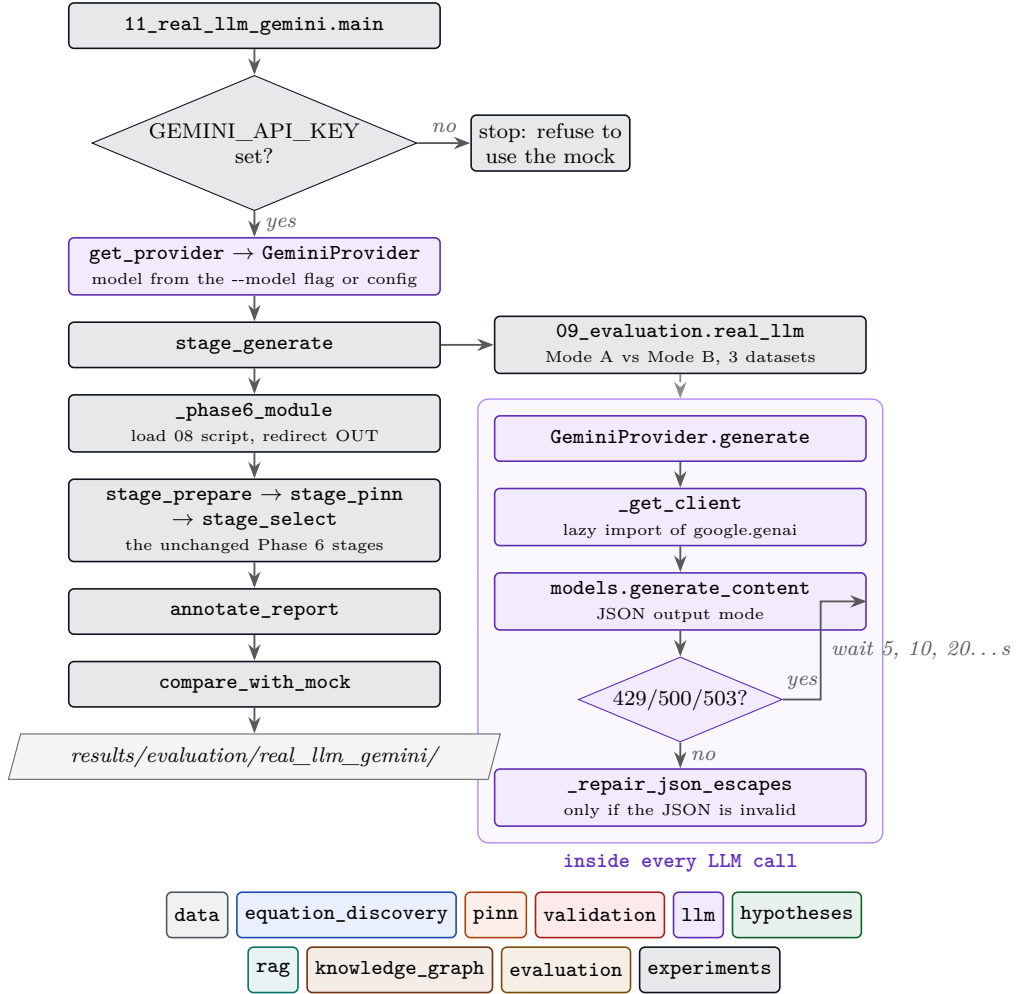


Figure 10: Real-LLM (Gemini) run and the provider's retry/JSON handling.